KARTHIK A

SENIOR CLOUD OPS ENGINEER

#74/1, 7th main, Vinayak Nagar, Mysuru 570012 | karthikag021@gmail.com | +91 8553833453

PROFESSIONAL SUMMARY

Results-driven DevOps and Cloud-Native Engineer with 10+ years of experience designing and operating scalable, secure, and highly available cloud platforms. Specialized in Kubernetes, Azure cloud services, Azure DevOps, and GitOps using ArgoCD to enable reliable, automated deployments. Strong expertise in observability and monitoring with the LGTM stack, and in integrating DevSecOps practices to enhance security, compliance, and operational efficiency. Actively contributing to AI infrastructure initiatives, integrating Azure OpenAI, OCR, Document Intelligence, semantic routing, and content safety solutions, including hosting and optimizing Ollama workloads on GPU-enabled Kubernetes clusters.

CORE COMPETENCIES

Cloud & Infrastructure

- Cloud Platforms: Microsoft Azure (AKS, Virtual Machines, Azure Functions), Kubernetes (k8s, k3s), Rancher, Edge computing architectures
- Infrastructure as Code: Terraform, Pulumi, Azure ARM Templates, Helm; GitOps implementation using ArgoCD
- Container & Platform Engineering: Docker, Kubernetes, Service Mesh (Istio, Gloo Mesh), distributed storage solutions including Longhorn and Rook
- Hybrid & Multi-Cloud Architecture: Design and management of secure, compliant, and scalable workloads across hybrid and multi-cloud environments

AI Engineering & LLM Infrastructure

- Azure AI Ecosystem: Azure OpenAI Service, Azure AI Studio, OCR, Document Intelligence, Language Services, and Speech Services for enterprise-grade AI solution development
- Open-Source LLM Platforms: Deployment and optimization of GPU-accelerated workloads using Ollama, LLaMA, and vLLM for scalable model inference
- AI Gateway & Orchestration: Implemented semantic routing, content safety controls, AI request orchestration, and intelligent load balancing to ensure secure and reliable model consumption
- AI Application Integration: Built Retrieval-Augmented Generation (RAG) pipelines leveraging prompt engineering, embeddings, and vector databases for contextual AI solutions
- GPU-Optimized Kubernetes Workloads: Designed and operated GPU-enabled Kubernetes clusters hosting Docling, Ollama, and 3D rendering services to deliver high-performance, low-latency AI and visualization workloads

Observability & Monitoring (LGTM Stack)

- Monitoring & Metrics: Implemented Prometheus, Grafana, Mimir, and Alertmanager to design scalable, high-availability monitoring architectures with proactive alerting
- Centralized Logging: Engineered log aggregation pipelines using Loki, Promtail, and Fluent Bit for efficient log ingestion, correlation, and troubleshooting
- Distributed Tracing: Integrated Tempo for end-to-end trace visibility across microservices, improving root cause analysis and performance tuning
- Site Reliability Engineering (SRE): Defined SLIs/SLOs, implemented error budgeting practices, led incident response processes, and reduced MTTR through data-driven reliability improvements

DevOps, CI/CD & Security

- CI/CD & GitOps: Designed and maintained automated pipelines using Azure DevOps, GitLab CI, and GitHub Actions; implemented GitOps workflows with ArgoCD to enable reliable, version-controlled Kubernetes deployments
- DevSecOps Automation: Integrated container scanning (Trivy), SAST (SonarQube), DAST (Invicti/Netsparker, OWASP ZAP, Burp Suite Enterprise), and Falco runtime security into CI/CD pipelines to enforce shift-left security practices
- Cloud & Container Security: Implemented image signing and vulnerability management using Azure Defender, Harbor Registry, Docker Notary, Snyk, and Mend to strengthen supply chain security
- Code Quality & Artifact Management: Enforced code quality gates with SonarQube and managed secure artifact repositories using Artifactory to ensure build integrity and compliance

Identity & Access Management

- IAM Platforms: Azure Entra ID (AD), Keycloak, SAML, OAuth2, Kerberos
- Security: MFA, SSO, RBAC, ABAC, Pod Security Policies, Network Policies

Platform Engineering & Developer Experience

- Developer Portal: Core member of the solutioning team managing an in-house internal developer portal serving 300+ users, responsible for overall SRE and DevOps operations supporting platform reliability and developer experience.
- Platform Tooling: Designed and developed platform engineering tooling, including self-service infrastructure provisioning, standardized CI/CD templates, and observability blueprints.
- Developer Experience: Reduced cognitive load for developers by providing golden paths, reusable templates, and opinionated deployment patterns; improved time-to-production and operational reliability.
- GitOps: ArgoCD, Argo Workflow, policy-as-code

Programming & Scripting

- Languages: Python, PowerShell, Bash/Shell, Go (foundational)
- Automation: Infrastructure automation, custom tooling, operational scripts
- Additional Technologies
- Databases: PostgreSQL, MongoDB, TimescaleDB, MySQL, PgVector, Qdrant.
- Messaging and queue: Kafka, RabbitMQ, Redis
- Load Balancers & Ingress: NGINX, Traefik

PROFESSIONAL CERTIFICATIONS

Certified Kubernetes Administrator (CKA)

KEY ACHIEVEMENTS

- Delivered a technical session at KubeCon 2025 (Hyderabad) on Edge Observability using the LGTM stack, showcasing scalable monitoring architecture for distributed edge environments
- Runner-up in enterprise AI Hackathon for designing a multi-modal AI agent capable of intelligent skill selection, dynamic routing, and context-aware response generation using LLM orchestration principles

PROFESSIONAL EXPERIENCE

Senior Cloud Ops Engineer

Kongsberg Digital India | July 2022 – Present

Client: Norway-based IoT high-technology systems company
Domains: IoT, Edge Computing, Industrial Automation, Real-time Data Processing

Key Responsibilities & Achievements:

- Designed and implemented hybrid cloud architectures enabling secure, compliant, and seamless integration between edge environments and Azure
- Automated infrastructure provisioning using Terraform and Pulumi, delivering repeatable, version-controlled deployments and reducing manual configuration.
- Led the transition from YAML-based pipelines to a GitOps workflow using ArgoCD, enhancing deployment efficiency and automation.
- Designed and implemented a comprehensive monitoring stack using Prometheus, Thanos, and Grafana to provide real-time insights into cloud applications, edge apps, and cluster metrics. Integrated Grafana OnCall and Alertmanager for proactive alerting and incident response.
- Developed a centralized logging solution with Loki and intuitive Grafana dashboards for improved observability.
- Built a local developer environment replicating cloud infrastructure using WSL, K3s, ArgoCD, and KubeVirt.
- Embedded DevSecOps practices by integrating SonarQube, Mend, Trivy, and GitLeaks for proactive security.
- Managed Azure cloud administration, security, and threat modeling using Microsoft Threat Modeling Tool.
- Optimized infrastructure costs by analyzing resource utilization across development and production environments, achieving a 30% cost reduction.
- Designed and operated production-grade Azure OpenAI infrastructure enabling secure, scalable LLM inference; implemented access controls, monitoring, cost governance, and compliance automation to optimize performance and spending.
- Enabled Retrieval-Augmented Generation (RAG) platforms by provisioning and managing vector databases, containerized LLM services, and scalable Kubernetes-based inference environments.
- Engineered and maintained GPU-optimized Ollama hosting infrastructure on dedicated GPU nodes, improving inference latency and maximizing throughput for high-demand AI workloads
- Automated and supported OCR and Azure Document Intelligence pipelines by provisioning cloud resources, configuring secure integrations, and implementing observability for reliable, large-scale document processing.
- Managed GPU-enabled Kubernetes clusters hosting Docling, Ollama, and 3D rendering workloads; implemented GPU scheduling, node autoscaling, resource optimization, and performance tuning to ensure high availability and low-latency execution.
- Led end-to-end deployment of enterprise product into customer on-premises Azure environments, ensuring secure architecture design, automated provisioning, and seamless production rollout
- Developed and deployed custom Prometheus exporters to collect and expose application-specific metrics, improving observability coverage and enabling proactive performance monitoring
- Engineered JWT-based authentication mechanism to securely integrate OpenCost with Prometheus/Mimir, enabling authenticated metric ingestion and cost visibility across Kubernetes workloads

Senior Software Engineer

Siemens Technology and Services | July 2018 – June 2022

Key Projects & Achievements:

- Designed and deployed end-to-end hosting solutions for web applications on Azure, leveraging services like Key Vault, App Services, Function Apps, Azure Application Gateway and Azure PSQL DB
- Automated infrastructure deployment using ARM templates, Terraform, and Azure modules.
- Streamlined microservices deployment in Azure Kubernetes Service AKS with Helm charts and containerized workflows
- Built fully automated CI/CD pipelines using Azure and GitLab, optimizing deployment efficiency.
- Responsible for creating installers using wix.

Technology Consultant

DXC Technology | December 2015 – June 2018

Key Projects & Achievements:

- Experienced in application packaging and virtualization using industry-standard tools like AdminStudio Repackager, InstallShield 2015, and App-V
- Repackaged legacy installers and In-House applications using AdminStudio Repackager.

EDUCATION

ATME College of Engineering, Mysuru **June 2011 - April 2015**

- Bachelor of Engineering in Electronics and Communication